Tamseel Fatima

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EDUCATION

Jan. 2022 - Present — Ph.D. candidate in Animal and Avian Sciences **(GPA 3.916/4.0)** — MD, United States
University of Maryland, College Park
- **Dissertation Project:** "*Role of iron and iron importer TfR1 in beiging of inguinal white adipose tissue (iWAT)"* (**Advisor:** Byung-Eun Kim, Ph.D.)

Sep 2016 - Sep 2020 — Bachelor of Science in Microbiology & Molecular Genetics **(GPA 3.86/4.0, GOLD MEDAL)** — Lahore, Pakistan
University of the Punjab, Lahore
- **Thesis:** "Effect of iron overload and HCV on hepatic and renal functioning in beta-thalassemia patients" (**Advisor:** Shahida Hasnain, Ph.D.)

RESEARCH EXPERIENCE

Jan 2022- Present — **Graduate Research Assistant** — MD, United States
University of Maryland-College Park
- Project Topic: Role of iron and iron importer TfR1 in beiging of iWAT
- Research work involves: mouse handling, adipocyte isolation, cell culture, microscopy and other molecular biology techniques

Mar 2021- Dec 2021 — **Research Assistant** — Lahore, Pakistan
National Centre of Excellence in Molecular Biology (NCEMB)
- Project Topic: Amniotic membrane embedded phytoextract primed stem cells construct for wound healing – Higher Education Commission, Pakistan's funded project
- Research work involved: phytoextract preparation, extract characterization, isolating Adipose Dervied-Mesencymal Stem Cells (ADMSCs), priming ADMSCs, cell assays and microscopy

Jul 2020- Aug 2020 — **Intern** — Lahore, Pakistan
Camcer Research Centre, University of the Punjab
- Performed the literature review of original research work Published between 2000-2020 in the field of lifestyle changes and cancer.
- Submitted and published a report titled "Lifestyle and cancer prevention" to University of the Punjab stakeholders .

Jul 2019- Jun 2020 — **Research Assistant** — Lahore, Pakistan
Sundas Foundation, Hematological disorders and blood transfusion Centre
- Projects: (a) Food source based iron chelation strategies to improve quality of life for beta thalassemia patients, and (b) evaluation of the effect of iron overload and hepatitis C on hepatic and renal functioning in beta-thalassemia patients
- Complete blood count, blood typing test, ferritin measurement, ELISA based assays for HCV, and HAV, and mutational screening of beta-thalassemia patients.

PUBLICATIONS AND PRESENTATIONS

Publications

- (In revision after PNAS review) First author of "Iron-Dependent Regulation of Adenylyl Cyclase 3 Governs the Thermogenic Activation of Adipocytes" from Byung-Eun Kim Lab.
- Fayyaz, T., Qadeer, M. I., Irfan, M., Amjad, F., **Fatima, T.,** Husnain, S., (2023). Effect of Flaxseed intervention on iron overload and its complications in Thalassemia Major patients, Nutrition and food science, 8(53), 1268-1278.
- Amjad, F., **Fatima, T.,** Fayyaz, T., Khan, M. A., & Qadeer, M. I., (2020). Novel genetic therapeutic approaches for modulating the severity of β-thalassemia. Biomedical Reports, 13(5), 48.

Conference Presentations:

- **Tamseel Fatima**, Yasmin A. Mejia-Guevara, Tamara Korolnek, Evan Ying, Yining Liu, Theresa A. Alexandar, Naili Liu, Najib M. El-Sayed, Oksana Gavrilova and Byung-Eun Kim, "Insufficient basal iron limits beiging via alternative Adenylyl cyclase 3 transcription in adipocytes," Red Cell Club (RCC) and East-West Iron Club (EWIC) meeting, School of medicine, University of Maryland, Baltimore, MD, USA, October 27-29, **Oral presentation**
- **Tamseel Fatima**, Yasmin Mejia-Guevara, Evan Ying, Yinning Liu, Najib El-Sayed, Oksana Gavrilova, and Byung-Eun Kim, "Adipocyte iron level orchestrates cAMP driven browning of white adipose tissue," Mid-Atlantic Frontiers in Metals and Medicine Symposium, University of Maryland, MD, USA, September 25, 2024, **Poster presentation**
- **Tamseel Fatima**, Yasmin Mejia-Guevara, Evan Ying, Yinning Liu, Najib El-Sayed Oksana Gavrilova, and Byung-Eun Kim, "Iron modulates cyclic AMP levels to promote the browning of white adipose tissue", Annual Mid-Atlantic Diabetes and Obesity Research Symposium, Diabetes, Endocrinology, and Obesity Branch (DEOB), National Institutes of Health (NIH), MD, USA, September 27, 2024, **Flash talk** and **Poster presentation**
- **Tamseel Fatima,** Yasmin Mejia-Guevara, Evan Ying, Yinning Liu, Oksana Gavrilova, and Byung-Eun Kim, "Iron modulates cyclic AMP levels to promote the browning of white adipose tissue", 2024 FASEB Meeting - Trace Elements in Biology and Medicine, Melbourne, FL, USA, June 9, 2024, **Flash talk and poster presentation**
- **Tamseel Fatima,** Yasmin Mejia-Guevara, Evan Ying, Oksana Gavrilova, and Byung-Eun Kim, "Iron is a key modulator for beiging of inguinal white adipose tissue (iWAT): implications for metabolic health", Annual Mid-Atlantic Diabetes and Obesity Research Symposium, Diabetes, Endocrinology, and Obesity Branch (DEOB), National Institutes of Health (NIH), MD, USA, September 22, 2023, **Poster presentation**

Departmental Presentations (ANSC Annual symposium in University of Maryland, College Park, MD, USA):

- **Tamseel Fatima,** Yining Liu, Ashton Trey Belew, Evan Ying, Najib El-Sayed, Byung-Eun Kim, "Transferrin Receptor 1 is essential for early postnatal adipose tissue development and remodeling," May 2025, **Poster presentation**
- **Tamseel Fatima,** Yasmin Mejia-Guevara, Evan Ying, Yinning Liu, Oksana Gavrilova, and Byung-Eun Kim, "Iron modulates cAMP levels to promote beiging of inguinal white adipose tissue", May 23, 2024, **Oral presentation**
- **Tamseel Fatima,** Yasmin Mejia-Guevara, Evan Ying, and Byung-Eun Kim, "Iron is a key modulator for thermogenic response and beiging in inguinal white adipose tissue (iWAT): implications for metabolic health", May 25, 2023, **Poster presentation**
- **Tamseel Fatima,** Yasmin Mejia-Guevara, Evan Ying, and Byung-Eun Kim, "Metabolic crosstalk between iron homeostasis and UCP1 regulation in adipose tissue", May 25, 2022, **Poster presentation**

GRANT WRITING EPXPERIENCE AND ACADEMIC SERVICES

Grants

- Secured **MPower Early Scholar Investment Fund** to support summer and fall 2026 stipend, tuition fee and fringe benefits (January 2026).
- Contributed to the design and writing of a specific aim for an **NIH R21** grant proposal titled *"Fe-Driven Genetic and Metabolic Programming of Adipocytes".* The application was submitted by PI Byung-Eun Kim, Ph.D. (October 2025)
- Contributed as a Co-Investigator (Co-I) to writing portions of a grant submission to the Maryland Agricultural Experiment Station **(MAES)** Competitive Grant Program. The proposal titled *"The Role of Iron in Adipocyte Signaling and Homeostasis,"* was awarded to **PI** Byung-Eun Kim, Ph.D. (November 2024)
- Contributed to the design of experiments and review of the final draft of an **NIH R03** grant proposal titled *"Iron Import Dynamics: Implications for Adipose Tissue Development and Homeostasis".* The application was submitted by PI Byung-Eun Kim, Ph.D. (June 2024)

Manuscript review

- Co-reviewer (Early Career Researcher) for a manuscript submitted titled "*Pathophysiological Significance of Impaired KAT7-Dependent Histone H3K14 Acetylation During Zinc Deficiency"* to *Nature Communications (2025)*.

Training and mentoring experience

- Served as co-advisor for **undergraduate honors thesis research** (2023-2026) of Yining Sunnythia Liu, *"Investigating the Mobilization of Hepatic Iron During Early Development of Mice."* She successfully

presented her research on March 27, 2026. Primary Advisor: Byung-Eun Kim, Ph.D., Committee members: Younggeon Jin, Ph.D. and Andrew Schiffmacher, Ph.D.

TEACHING EXPERIENCE

Spring 2026 & Spring 2024 — **Teaching Assistant for Principles of Animal Science (ANSC 101)** — MD, United States
University of Maryland-College Park

- Graded assignments and provided feedback to students (50-60 students)
- Delivered 2 guest lectures on The Science of Animal Science: 1. Navigating and Understanding the Research Article, and 2. Interpreting Data in Research Articles and Evaluating Research Qualtity
- Held discussion and term-review sessions for students
- Prepared comprehensive course review sheets for students
- Managed course website, including lecture materials, announcements, and grades

Fall 2025 — **Guest lecturer for Nutritional aspects of metabolic disorders (ANSC454/688O)** — MD, United States
University of Maryland-College Park

- Guest lecture on topic: Fe homeostasis in animals

Spring 2025 — **Teaching Assistant for Applied Animal Physiology Lab (ANSC 214)** — MD, United States
University of Maryland-College Park

- Assisted in lab instruction, experiment setup, and student support during weekly sessions (40 students)
- Independently led lab section, delivered lectures, guided experiments, and graded assignments (20 students)
- Managed course website, including lecture materials, announcements, and grades

Fall 2022 — **Teaching Assistant for Eating with Eyes Wide Open (ANSC 227)** — MD, United States
University of Maryland-College Park

- Graded assignments and journal entries (59 students)
- Provided feedback to students on their journal entries to help improve there writing and analytical abilities
- Guest lecure on gut microbiome and animal health aspects

SCHOLARSHIPS, FELLOWSHIPS AND AWARDS

- **Best Oral Presentation Award** for short talk at Red Cell Club (RCC) and East West Iron Club (EWIC) meeting at School of medicine, University of Maryland, Baltimore, MD, USA, October, 2025.
- **ARPAS-ANSC Travel Award** for presenting at 2024 FASEB Meeting - Trace Elements in Biology and Medicine, Melbourne, FL, USA, April, 2024.
- **Summer Research Fellowship** from UMD Graduate School for carrying out summer research project "Iron boosts a secondary messenger cAMP to burn the white fat", March 2024.
- **Dean's Fellowship Award** from UMD Graduate School, awarded to outstanding graduate students, October 2022.
- **Undergraduate Poster Presentation (First prize),** CancerBio conference, Department of Microbiology and Molecular Genetics, University of the Punjab, Lahore, Pakistan, Feb 19, 2020.
- **Undergraduate Merit Scholarship**, Department of Microbiology and Molecular Genetic, University of the Punjab, Lahore, Pakistan, Sep 2019.
- **Scientific Essay Writing Award (Second prize),** SUNMAC, Sundas Foundation, Lahore, Pakistan, Sep 2019.
- **Undergraduate Merit Scholarship**, Department of Microbiology and Molecular Genetic, University of the Punjab, Lahore, Pakistan, Sep 2018.
- **Poster Presentation Award (First prize),** Department of Microbiology and Molecular Genetic, University of the Punjab, Lahore, Pakistan, Mar 2018.
- **Scientific Essay Writing Award, (Second prize),** Astrobiology Network of Pakistan, Lahore, Pakistan, Aug 2018.
- **Undergraduate Merit Scholarship**, Department of Microbiology and Molecular Genetic, University of the Punjab, Lahore, Pakistan, Sep 2017.

- **Primary School Distinction Merit Scholarship**, Government of Punjab, Pakistan, July 2013.

TECHNICAL SKILLS

- Microbiology techniques: bacterial culture, enumeration of bacteria, identification of bacterial strains, antibiotic sensitivity testing, agar well diffusion assay, bacterial growth curve determination, bacteriophage isolation, plaque assay
- Animal experiments: Mouse handling, ear tagging, toe clipping, IP, Sub-Q injections in C57BL/6J mice
- Cell culture: primary cells (human and mouse adipocytes) isolation and cell line generation, Assays for cell viability and proliferation, cell-cycle analysis
- Molecular biology techniques: cloning, transformation, transfection, transduction, DNA and RNA isolation, genotyping/SNP genotyping (PCR based), PCR, qPCR, digital PCR, Protein extraction and quantification, SDS-PAGE, western blotting
- Histology and imaging: cryotome and microtome, H&E and picrosirius red staining, immunofluorescence staining, confocal, fluorescence and light microscopic imaging
- ELISA assays for virus proteins and cyclic AMP
- RNA sequencing and bioinformatics analysis: Bulk RNA sequencing, Analysis of RNA-sequencing, single cell RNA-sequencing datasets using hpgltools, and Seurat on R studio respectively.
- Software and tools for analysis of biological information: Alpha fold, NCBI, UCSC genome browser, promoter analysis, primer3, snapgene, Rstudio.

SCIENTIFIC CONTENT WRITING EXPERIENCE

Dec 2019-Jan 2022 — **Scientific Content Writer (Remote)** — Ebersberg, Germany
Eurofins Genomics
- Wrote scientific articles related to biotechnology, molecular biology, and diagnostics under the supervision of Dr. Andreas Ebertz
- Wrote introductory articles for products launched by Eurofins Genomics

Sep 2020-Oct 2020 — **Scientific Content Writer** — Islamabad, Pakistan
Pakistan Aging Research Society
- Organized events to raise awareness about aging in general public
- Presented theories of aging and recent advancements in the field of anti-aging
- Wrote blog articles on challenges faced by aging population

Summer 2020 — **Intern (Writing Track)** — Pakistan
Spectra Online Science Magazine
- Interviewed Enterpreneur Emily Hicks about her start-up FREDsense and published it
- Learnt communicating science to masses by getting training in the scientific writing track

Spring 2018 — **Guest Blog Writer** — Pakistan
Bioscience Pakistan
- Wrote guest blog articles, managed, and published them on website

SCIENCE OUTREACH AND BLOG ARTICLES

- Tamseel Fatima and Andreas Ebertz, Microarrays Accelerate Blue Biotechnology to The Next Stage, Sep 2021, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz, The Future of RNA Applications: Diagnostics, Therapeutics and Research, Apr 2021, The DNA Universe blog.
- Tamseel Fatima, Girl's and Boy's Day 2021: Become The Next Generation of Biologists, Apr 2021, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz, International Women's Day 2021. We Are Celebrating The Women In Biology, Mar 2021, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz, The history of mRNA applications, Apr 2021, The DNA Universe blog.

- Tamseel Fatima, What Why and How of mRNA Vaccines, Feb 2021, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz, A Journey Through the History of DNA Sequencing, Nov 2020, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz, Who Won the Race to Solve the DNA Structure? June 2020, The DNA Universe blog.
- Andreas Ebertz and Tamseel Fatima, SARS-CoV-2 Genomics Facts, May 2020, The DNA Universe blog.
- Tamseel Fatima and Andreas Ebertz ,Why You Need to Authenticate Your Cell Line – Your Research Could Depend on It, Mar 2020, The DNA Universe blog.
- Tamseel Fatima, Gut feeling induced by gut flora, March, 2018, BioScience.pk.

OTHER EXPERIENCES

2024-2025 **ASGSA (Animal Science Graduate Student Association) vice president** MD, United States
University of Maryland-College Park
- Pariticipated in quarterly meetings to kick start the organization
- Organized and participated in events for ANSC graduate community

2023-2024 **ANSC DEIR (Diversity, Equity, Inclusion, Respect) member** MD, United States
University of Maryland-College Park
- Part of the DEIR sub-committee to promote DEIR practices for recruiting, retaining, and developing diversity
- Gathered information about scholarships and fellowships that promote DEIR
- Contributed to poster design for DEIR efforts and presented it at annual ANSC symposium 2024.